

Import Duty Pass-Through to Domestic Gold Prices: Quasi-Experimental Evidence from India's Gold Market, 2024–2026

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Abstract

This paper estimates how much of India's May 2026 gold import duty hike, from 6 to 15%, passed through to domestic gold prices. Exploiting the sharp and unanticipated policy discontinuity as a quasi-experiment, this paper applies an interrupted time series (ITS) design with Newey-West HAC standard errors to a primary sample of 338 daily observations from the India Bullion and Jewellers Association benchmark price series (July 2024 – July 2026), with 307 pre-hike and 31 post-hike trading days. The full dataset comprises 826 IBJA trading days spanning January 2022 through July 2026, within a 1,172-row weekday date spine through July 2, 2026.

The duty hike raised the domestic gold premium by Rs. 10,136 per 10 grams, representing 84.5% of the theoretical ceiling, with a 95% confidence interval of 77.3 to 91.8%. Full pass-through is formally rejected ($t = -4.16$, $p < 0.001$). Local projections confirm the adjustment was immediate on the first trading day and held on a stable plateau through Day 14 (77.5%), with no evidence of anticipatory pricing. An ARDL bounds test confirms cointegration between domestic and international gold prices in the pre-hike period; the stable local-projection plateau through Day 14 is consistent with the shift persisting rather than mean-reverting within the observation window. Three consistency checks using counterfactual methods trained on the pre-hike window converge within 5% of the primary estimate, and fake-date placebo tests confirm the result is specific to May 13, 2026.

This paper also documents asymmetric price transmission: the duty hike passed through at 84.5% by the first trading day, while the July 2024 cut reached only 70.8% at an equivalent horizon (clean sample, July 2022–May 2026, $N = 687$). The hike therefore transmitted more completely than the cut, consistent with downward rigidity in duty pass-through. The incomplete pass-through is consistent with supply-side dampening from front-loaded inventory imported at the prior duty rate, seasonal demand weakness, and historical smuggling elasticity.

*Data and replication code are available at <https://github.com/OlixIgnacious/gold-policy-project>. The dataset contains 826 daily IBJA price observations spanning January 2022 – July 2026, drawn from 826 IBJA daily PDF circulars, within a 1,172-row weekday date spine (January 3, 2022 through July 2, 2026) that also includes 346 rows with no price observation (236 official market holiday and closure days and 110 IBJA archive gap dates).

Keywords: import duty pass-through, gold market, India, interrupted time series, quasi-experiment, commodity taxation

JEL Codes: E31, F13, H22, Q02

1 Introduction

India is the world's second largest consumer of gold, importing between 700 and 900 tonnes annually and accounting for roughly a quarter of global physical demand^[1]. Gold imports are simultaneously a major source of current account pressure: in fiscal year 2026, gold import spending reached \$71.98 billion, approximately 9% of total merchandise imports, contributing to a current account deficit of \$30.1 billion between April and December 2025. In this context, the import duty on gold is not merely a revenue instrument. It is an active lever of macroeconomic management, deployed periodically to compress import demand and stabilise the rupee.

On July 23, 2024, the Union Budget reduced the basic customs duty on gold from 10% to 5% and the Agriculture Infrastructure and Development Cess (AIDC) from 5% to 1%, lowering the combined duty from 15 to 6%. This was the lowest duty rate on gold in over a decade. The immediate response was a sharp compression of domestic gold premiums toward import parity and a surge in demand, with the World Gold Council estimating the cut would add at least 50 tonnes to second-half 2024 demand^[2]. FY26 gold imports rose 24% year-on-year versus FY25. By early 2026, with the rupee under sustained depreciation pressure and a persistent current account deficit, the government acted again. On the evening of May 12, 2026, a gazette notification restored the combined duty to 15%, effective from midnight, reversing the 2024 cut in its entirety and taking full effect at the opening of markets on May 13, 2026.

This paper asks a precise question: how much

of the 9 percentage point duty increase passed through to the domestic gold price? The answer is not obvious. In a textbook small open economy with competitive markets and free trade, a duty increase of $\tau\%$ should raise the domestic price by exactly $\tau\%$ of the import parity price, yielding 100% pass-through. But India's gold market has structural features that push against complete pass-through. Smuggling, which historically rises sharply when domestic premiums approach the ceiling implied by full pass-through, provides a competing supply channel that caps price increases^[3]. Demand destruction compresses the premium from the demand side. And short-run inventory dynamics, particularly the large volume of gold imported at the prior 6% duty rate in April 2026 (\$5.6 billion), add a transitory supply buffer that suppresses the immediate price response. Measuring the actual pass-through, and understanding why it falls short of 100%, is the central empirical challenge this paper addresses.

This paper exploits the May 13, 2026 duty hike as a quasi-experiment. The hike was announced at midnight with no legislative process and no public pre-announcement, making it operationally equivalent to a sharp and unanticipated discontinuity in the policy environment. This paper applies an interrupted time series (ITS) design to a dataset drawn from the India Bullion and Jewellers Association (IBJA) benchmark price series spanning January 2022 through July 2026, comprising 826 daily price observations from IBJA PDF circulars. The outcome variable is the domestic gold premium: the gap between the IBJA published benchmark price and the import parity price implied by the 6% duty regime. By fixing the parity baseline at the pre-hike duty

rate, the premium directly measures how much of the duty shock has been absorbed into domestic prices.

The main finding is that the duty hike raised the domestic gold premium by Rs. 10,136 per 10 grams (Newey-West HAC standard error Rs. 445, $p < 0.001$), representing 84.5% of the theoretical duty ceiling. The null hypothesis of full pass-through is formally rejected ($t = -4.16$, $p < 0.001$). Local projection estimates^[4] confirm that adjustment was immediate and held on a stable plateau through Day 14 (77.5%), with no anticipatory pricing in the pre-hike window. An ARDL bounds test^[5] confirms the pre-hike cointegrating relationship between domestic and international gold prices; the stable local-projection plateau through Day 14 is consistent with the shift persisting rather than mean-reverting within the observation window. Three consistency checks using counterfactual machine-learning methods trained on the pre-hike window converge within 5% of this estimate, and fake-date placebo tests confirm the result is specific to May 13, 2026. Full results are reported in Section 6.

This paper makes three contributions to the literature on commodity tax incidence and price transmission. First, it provides the first causal daily-frequency estimate of import duty pass-through in India's gold market, using a quasi-experimental design that cleanly separates the duty effect from concurrent movements in the international gold price and the exchange rate. Prior work on Indian gold prices has been largely descriptive or confined to monthly frequency, precluding identification of the immediate price response. Second, it documents asymmetric price transmission in this market. Comparing the May 2026 hike with the July 2024 cut, this paper finds that the duty hike passed through more completely (84.5%) than the July 2024 duty cut (70.8%), with both episodes registering immediate adjustment at Day 0. This asymmetry – duty increases transmitting more completely than duty reductions – has direct implications for the de-

sign of duty policy as a demand management tool: hikes reach the market rapidly and nearly in full, while cuts are partly absorbed by dealers holding high-duty inventory who resist the full price reduction. Third, by combining the ITS estimate with local projections, GARCH volatility analysis, and news sentiment classification, this paper provides a multi-method characterisation of the market adjustment process that goes beyond a single point estimate.

The remainder of this paper is organised as follows. Section 2 reviews the related literature. Section 3 describes the policy background. Section 4 introduces the data and the construction of the import parity and domestic premium measures. Section 5 presents the empirical strategy. Section 6 reports the main results. Section 7 presents robustness checks. Section 8 discusses the partial pass-through mechanisms. Section 9 concludes.

2 Literature Review

This paper connects three strands of the literature: the economics of tariff and duty pass-through, the Indian gold market, and the local projection and interrupted time series methodologies.

Tariff and Import Duty Pass-Through

The foundational question of whether trade barriers are transmitted to domestic prices has a long lineage in international economics. Menon^[6] surveys the exchange rate pass-through literature, finding that commodity markets exhibit substantially higher pass-through than differentiated manufactures, a pattern attributed to the absence of local pricing power and the thinness of distribution margins. Campa and Goldberg^[7] extend this to import prices specifically, documenting that pass-through is typically incomplete in the short run and that commodity sectors – particularly metals and raw materials – approach full pass-through more rapidly than finished goods.

The literature on tariff incidence took on renewed empirical urgency following the 2018–2019 US–China trade escalation. Fajgelbaum et al.^[8] exploit cross-product variation in tariff schedules using a difference-in-differences design and find that US tariffs were almost entirely borne by domestic importers and consumers, with the terms-of-trade effect predicted by theory failing to materialise at the observed scale. Their finding of near-complete tariff pass-through in commodity-like product categories is the closest analog to the gold market setting studied here. Gopinath et al.^[9] show that the currency of invoicing is a key determinant of pass-through completeness; Amiti et al.^[10] further document that high-market-power producers absorb cost shocks in their mark-ups rather than passing them through to buyers. The present paper documents precisely such a wedge – a 15.5% absorption gap – and traces it to supply-side structural features specific to the Indian gold market rather than to invoicing currency or seller market power.

Indian Gold Market

India’s gold market has attracted growing attention from both policymakers and researchers, reflecting the metal’s unusual position as simultaneously a consumption good, a savings vehicle, and a current account pressure point. Kanjilal and Ghosh^[11] estimate the price elasticity of gold import demand in India at between -0.69 and -1.01 , indicating that price increases from duty hikes should compress demand substantially but not eliminate it. This elasticity range provides a structural benchmark for back-calculating how much of the post-hike absorption gap could be attributed to demand compression, even in the absence of direct quantity data.

Maria Immanuvel Susai and Lazar Daniel^[3] document the relationship between India’s statutory duty rate and unofficial gold imports, establishing via Granger causality tests that customs duty encourages smuggling activity in the long run. Their analysis of seizure data shows that

informal imports are concentrated at major international air gateways—Dubai and the Gulf states supply the primary channel, with entry concentrated at Kerala, Chennai, and Mumbai airports—and peak in the October–December festive season. This supply-side substitution mechanism is central to understanding why pass-through in India’s gold market is structurally incomplete: the shadow supply curve is elastic at the margin, capping the premium that the formal market can sustain. World Gold Council data corroborate this channel: excluding the COVID years of 2020–21, the correlation between India’s statutory duty rate and unofficial imports is $+0.52$ ^[12].

Identification: ITS, Local Projections, and ARDL

Exploiting a sharp, unanticipated policy discontinuity as a natural experiment is a well-established identification strategy in empirical economics^[13]; interrupted time series is a tractable implementation for high-frequency continuous outcomes^[14]. The key identification assumption – that the outcome series would have followed the pre-treatment trend absent the intervention – is testable via placebo analysis and pre-trend tests, both of which this paper reports.

The local projection estimator introduced by Jordà^[4] provides a flexible, non-parametric alternative to vector autoregression for tracing the dynamic response of an outcome variable to a shock at multiple horizons. Unlike a VAR, local projections do not impose cross-horizon restrictions and are robust to model misspecification. Their use for tariff and duty pass-through analysis has been formalised in recent Federal Reserve research on the 2025 US tariff episodes, where the method’s ability to distinguish immediate from gradual adjustment proves particularly valuable. The present paper adopts this approach to document that gold price adjustment was immediate rather than gradual, with the premium plateau stabilising within the first trading day.

The ARDL bounds test of Pesaran et al.^[5] provides a third identification angle, testing whether the structural break constitutes a permanent shift in the long-run equilibrium between domestic and international prices or a transitory deviation that will mean-revert. The bounds test is particularly well-suited to financial series that may be a mix of I(0) and I(1) processes, and its F-statistic provides a clean test of cointegration without requiring unit root pre-testing. Heteroskedasticity and autocorrelation-consistent standard errors throughout follow Newey and West^[15].

3 Policy Background

India's gold import duty has served as an active macroeconomic instrument since at least 2013, when the government raised the combined duty in steps from 4 to 10% in response to a widening current account deficit and rupee depreciation, with a further increase to 12.5% in July 2019, a brief reduction to 10.75% in February 2021, and a restoration to 15% in July 2022. That episode established the template that would be repeated in 2026: duty hikes are deployed as a demand compression tool when gold imports threaten to widen the current account, and duty cuts are deployed as a stimulus tool when the government wishes to formalise demand and reduce smuggling incentives.

The July 2024 Duty Cut

On July 23, 2024, the Union Budget reduced the basic customs duty on gold from 10% to 5% and the Agriculture Infrastructure and Development Cess from 5% to 1%, lowering the combined effective duty from 15 to 6%. This was the lowest duty rate on gold since the 2013 escalation cycle and was framed explicitly as a measure to curb smuggling by narrowing the price differential between the formal and informal markets^[2].

The market response was immediate and substantial. The domestic premium collapsed toward import parity within the first week, and the World Gold Council estimated that the cut would add at least 50 tonnes to second-half 2024 jewellery demand^[16]. Between the cut and the subsequent hike, FY26 gold imports rose 24% year-on-year versus FY25, with total gold import spending in fiscal year 2026 reaching \$71.98 billion, approximately 9% of total merchandise imports. The current account deficit stood at \$30.1 billion between April and December 2025, narrowing from \$36.6 billion in the same period a year earlier.

Pre-Hike Supply Disruptions (March–May 2026)

The weeks before the May 2026 hike were characterised by multiple independent supply shocks that had already begun tightening the domestic gold market before the duty announcement. These disruptions are material for the empirical design because they created a departure from the clean low-duty baseline that the ITS framework assumes.

First, in March 2026, the Iran–US conflict disrupted commercial air routes from the UAE – a key transit hub for India's gold imports, accounting for approximately 24% of imports in 2025. Domestic market discounts widened to a monthly average of \$46 per troy ounce against London Fix, driven primarily by demand softness as international gold prices fell approximately 12% during the month; the supply disruption further reduced formal import volumes, with March 2026 imports falling to a 9-month low of approximately US\$3.1 billion^[17]. Second, from April 1, 2026, banks and bullion dealers suspended gold imports following uncertainty over IGST (Integrated Goods and Services Tax) exemption enforcement; seventeen banks were subsequently re-authorised by DGFT to resume imports on April 17, with full resumption occurring around May 12, further constraining formal supply in

the interim. Third, on April 2, 2026, the government imposed administrative restrictions on imports of gold and silver jewellery via the UAE free trade agreement route, which had been used to circumvent the standard duty structure. These three disruptions collectively produced a pre-hike tightening that pushed discounts from \$46 per ounce in March to approximately \$8 per ounce in early April. Fourth, and most significant for the absorption analysis, importers and refiners massively front-loaded purchases around the Akshaya Tritiya festival on April 19–20. April 2026 imports reached US\$5.6 billion, up more than 80% year-on-year, at the prior 6% duty rate^[12].

The May 13 Notification

On the evening of May 12, 2026, the Ministry of Finance issued Gazette Notification No. 15/2026-Customs and No. 16/2026-Customs, effective from midnight, restoring the basic customs duty on gold to 10% and the AIDC to 5%, returning the combined duty rate to 15% in its entirety. The rupee had closed that day at Rs. 95.63 per US dollar, its weakest level on record, reflecting sustained capital outflow pressure and elevated gold import demand. The notification was issued without parliamentary process, relying instead on the emergency customs notification authority available to the central government. The specific duty instrument and its timing were not publicly announced in advance; on May 11, 2026, the Prime Minister publicly urged families to avoid purchasing gold, a general appeal that did not reveal the duty mechanism or implementation date^[12]. The anticipation test in Section 7 (dropping the five pre-hike trading days) finds a change in $\hat{\beta}_1$ of -0.3% , providing no evidence that this public appeal caused anticipatory price movements. Indian markets opened on May 13 with the new duty structure fully in force.

4 Data

Primary Price Series

The primary price data come from the India Bullion and Jewellers Association (IBJA), which publishes daily benchmark rates for gold (999 purity) in rupees per 10 grams for both morning and afternoon sessions. The IBJA benchmark is the reference rate used by organised jewellers, bullion dealers, and commodity exchanges for physical transactions. It is set each trading day through a process involving major member firms and reflects the domestic wholesale clearing price inclusive of applicable duties and taxes. The afternoon session rate (PM fix) is used throughout, as it incorporates same-day duty and exchange rate information and is the more widely referenced settlement price.

IBJA data were collected directly from daily PDF circulars published on the IBJA member portal, with 826 PDFs scraped covering the period January 3, 2022 through July 2, 2026, all successfully parsed, yielding 826 PM gold price observations. A further 346 weekday dates in the dataset carry no price observation: 236 are official IBJA market holiday and closure days, and 110 fall within four identified archive gap clusters (January–April 2022, July–September 2023, August–October 2024, and October 2025) – dates on which IBJA is believed to have published prices but the PDF is not available in our archive, representing potential non-random missingness bounded in scope. BullionWorld.in, a licensed IBJA data republisher, was evaluated as a supplementary source for these 110 archive-gap dates and 47 additional holiday dates (157 total); 105 returned real contemporaneous prices (Rs. 47,583–Rs. 127,247 per 10g) and are used in a source robustness check reported in Section 7; the remaining 52 returned a placeholder value indicating no data available. No BullionWorld data are used in the primary dataset. All rows without a price are excluded from all regressions. The provenance of each observation is tracked

in the `ibja_source` column (pdf: 826 rows; NaN: 346 rows). The 826 usable PDF price observations constitute the observation pool for all analysis. The full dataset with source flags is publicly available at the replication repository.¹

International gold prices (USD per troy ounce) and the INR/USD exchange rate are sourced from Yahoo Finance, with daily close prices used for both series.

Import Parity and Premium Construction

The theoretical import parity price at the 6% duty rate – the regime prevailing between July 24, 2024 and May 12, 2026 – is constructed as:

$$P_t^{\text{parity}} = \text{Gold_USD}_t \times \frac{10}{31.1035} \times \text{FX}_t \times 1.06 \quad (1)$$

where $10/31.1035$ converts from USD per troy ounce to USD per 10 grams, and $\times 1.06$ applies the combined duty. The post-hike theoretical ceiling – the parity price at the 15% duty rate – is computed analogously with the multiplier

1.15. The domestic gold premium, the primary outcome variable, is defined as:

$$\text{Premium}_t = \text{IBJA_PM}_t - P_t^{\text{parity}} \quad (2)$$

By fixing the parity baseline at the pre-hike 6% duty rate throughout the sample, the premium is interpretable as the cumulative market absorption of the duty change: a premium equal to the full ceiling (1 with multiplier 1.15 minus multiplier 1.06) would indicate 100% pass-through. This construction ensures that movements in international gold prices and the exchange rate are netted out of the outcome variable mechanically rather than only through regression controls.

As a validation check, the mean domestic premium during the 6% duty regime (July 24, 2024 – May 12, 2026) is Rs. –257 per 10 grams across 321 IBJA trading days. While a one-sample t -test rejects a mean of exactly zero ($p = 0.001$), the deviation represents less than 0.4% of the prevailing domestic gold price (approximately Rs. 72,000 per 10 grams over this window), confirming that the parity formula is correctly calibrated to within economically negligible precision.

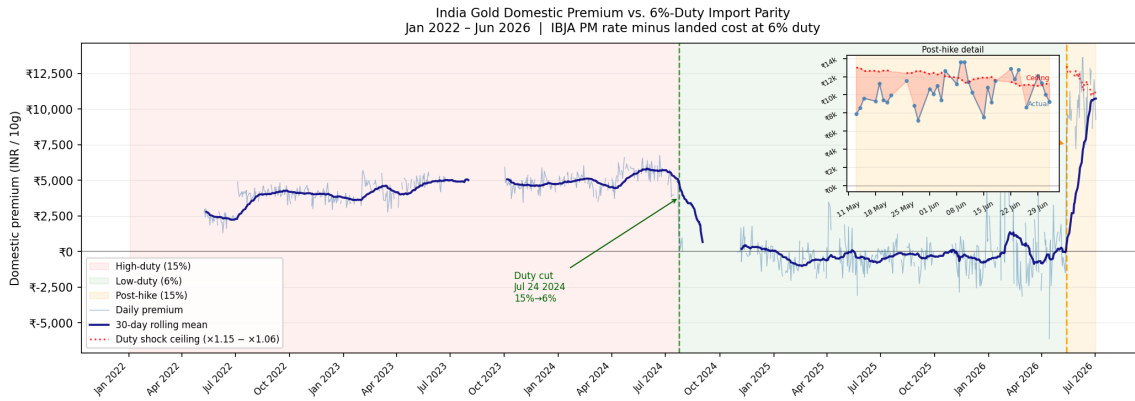


Figure 1: Domestic gold premium (IBJA benchmark minus import parity at 6% duty), January 2022 – July 2, 2026. The premium is near zero throughout the low-duty window (July 2024 – May 2026) and jumps sharply at the May 13, 2026 duty hike. Shaded band marks the post-hike window.

¹See <https://github.com/OlixIgnacious/gold-policy-project>. The replication dataset is `gold_policy_clean.csv`.

Sample and Summary Statistics

The full date spine spans January 3, 2022 through July 2, 2026, covering 1,172 weekday rows. Of these, 826 carry an actual IBJA price (all sourced from PDF circulars); the remaining 346 rows are official IBJA market holidays, closure days, or archive gap dates, and are excluded from all regressions. The primary estimation sample, following the ITS design, restricts to the low-duty window (July 24, 2024 onward), yielding 338 observations with complete data across all outcome

and control variables, with 307 pre-treatment and 31 post-treatment days.

Table 1 reports summary statistics for the key variables across the pre- and post-hike sub-periods. The domestic premium shifts from a mean of Rs. -259 before the hike² to a mean of Rs. 10,321 per 10 grams in the post-hike window, against a theoretical ceiling of Rs. 11,989. The daily first difference of international gold prices ($\Delta \text{Gold_USD}$) and the exchange rate (ΔFX) serve as the primary controls in the regression models.

Table 1: Summary Statistics

Variable	Pre-hike	Post-hike
Domestic premium (Rs./10g)	-259	10,321
Theoretical ceiling (Rs./10g)	$-$	11,989
$\Delta \text{Gold_USD}$ (%)	0.12	-0.35
ΔFX (Rs./USD)	0.02	0.04
Observations	307	31

5 Empirical Strategy

Interrupted Time Series Design

The May 13, 2026 duty hike provides a clean quasi-experiment: it was announced at midnight without legislative process and without advance disclosure of the specific duty instrument or timing, making it operationally equivalent to a sharp and unanticipated discontinuity in the policy environment. This setting supports an interrupted time series (ITS) design, in which the treatment effect is identified from the discrete break in the outcome series at the policy date, controlling for its pre-treatment trajectory and for concurrent movements in global determinants of the domestic price.

The primary regression specification is:

$$\text{Premium}_t = \alpha + \beta_1 \cdot \text{Post}_t + \beta_2 \cdot t + \beta_3 \cdot \Delta \text{Gold_USD}_t + \beta_4 \cdot \Delta \text{FX}_t + \varepsilon_t \quad (3)$$

where Post_t is an indicator equal to one for trading days on or after May 13, 2026; t is a linear time trend capturing any pre-existing drift in the premium during the low-duty window; $\Delta \text{Gold_USD}_t$ is the day-over-day change in the international gold price in USD; and ΔFX_t is the day-over-day change in the INR/USD exchange rate. The parity construction in equation (1) removes the *level* effect of global gold prices and the exchange rate from the premium mechanically, so $\Delta \text{Gold_USD}_t$ and ΔFX_t capture only the *incremental* daily noise not absorbed by the parity formula. Standard errors are computed using the Newey and West^[15] heteroskedasticity

²The validation in Section 4 reports Rs. -257 across all $N = 321$ pre-hike IBJA trading days with a valid premium; the Rs. -259 here reflects the subset with complete control variables (321 minus the 14 days lacking Yahoo Finance international price data), whose slightly different composition shifts the mean by Rs. 2 per 10 grams.

and autocorrelation-consistent (HAC) estimator with lag length $l = \lceil 0.75 \cdot T^{1/3} \rceil$, where $T = 338$ is the primary sample size, yielding $l = 6$.

The coefficient of interest is β_1 , which measures the level shift in the domestic gold premium attributable to the duty hike. The pass-through rate is then $\hat{\beta}_1 / \bar{C}$, where $\bar{C}$ is the mean theoretical ceiling computed across the 31 post-hike observations.

Identifying Assumptions

The ITS estimator requires two conditions. First, absent the duty hike, the premium would have continued along its pre-treatment trajectory – formally, $\mathbb{E}[\text{Premium}_t \mid \text{Post}_t = 0]$ is well approximated by $\hat{\alpha} + \hat{\beta}_2 \cdot t$. This is tested by the pre-trend test in Section 6, which finds a pre-treatment trend of Rs. 0.78 per day with $p = 0.35$, providing no evidence of pre-existing drift. Second, no contemporaneous shock other than the duty change produced a discrete break in the premium on May 13. The primary threat is the pre-hike supply disruption sequence documented in Section 3: the April 2 import restriction and the IGST pause both tightened domestic supply before May 13 and could have introduced a positive pre-trend. The April 2 restriction control in the robustness section directly tests this channel. Additionally, because the parity baseline is fixed at the 6% duty rate throughout, any supply disruptions that occurred at fixed international prices and exchange rates would appear as premium movements in the pre-hike period rather than as a confounded treatment effect.

Local Projection: Dynamic Event-Study Path

To characterise the *dynamic* adjustment path of the premium rather than the average level shift, this paper estimates local projections^[4] at each horizon $h = 0, 1, \dots, 30$ trading days post-hike. For each horizon, a separate OLS regression is

estimated:

$$\text{Premium}_{t+h} = \alpha_h + \beta_h \cdot \text{Post}_t + \gamma_h \cdot t + \delta_h \cdot \Delta \text{Gold_USD}_t + \zeta_h \cdot \Delta \text{FX}_t + \varepsilon_{t+h} \quad (4)$$

The sequence $\{\hat{\beta}_h\}_{h=0}^{30}$ traces the impulse response of the premium to the duty shock. This non-parametric approach is robust to model misspecification and avoids the dynamic restrictions imposed by VAR-based alternatives. Horizons $h \geq 31$ are outside the current observation window.

ARDL Bounds Test

The ARDL bounds test of Pesaran et al.^[5] is used to test whether the domestic gold price and the import parity price were cointegrated in the full pre-hike period, and whether that cointegrating relationship survived the duty hike. Cointegration pre-hike establishes that the domestic price was tethered to international prices through the import channel, validating the identifying assumption that the pre-treatment premium was mean-stationary. A failure of cointegration post-hike would indicate a permanent shift in the long-run equilibrium, consistent with the duty change having created a new price regime.

6 Results

Pre-Trend Test and Sample Validation

Before estimating the treatment effect, two pre-checks are reported. First, a regression of the domestic premium on a linear time trend during the low-duty pre-hike window (July 24, 2024 – May 12, 2026, $N = 321$ IBA trading days with a valid domestic premium; 14 of these lack the Yahoo Finance international gold price data required by the ITS regression controls, reducing the ITS pre-treatment count to $N = 307$) yields a slope of Rs. 0.78 per trading day with $p = 0.35$, providing no evidence of a pre-existing upward trend that could inflate the ITS estimate. Second,

the ADF unit root test on the premium series within the low-duty regime rejects a unit root at the 1% level ($p = 0.000$), confirming that the pre-hike premium is stationary around its near-zero mean. Non-stationarity in the full sample reflects the structural break at May 13 rather than a true unit root process.

Main ITS Result

Table 2 reports the ITS results across three progressively controlled specifications, all estimated on the primary sample ($N = 338$, July 24, 2024 – July 2, 2026).

Table 2: ITS Regression Results (Primary Sample)

	(1) Baseline	(2) +Controls	(3) +Trend
β_1 (Post)	10,580*** (266)	10,297*** (351)	10,136*** (445)
β_2 (Trend)			0.7 (1.0)
β_3 (Δ Gold)		−605*** (74)	−607*** (75)
β_4 (Δ FX)		102 (192)	103 (194)
Constant	−259 (164)	−162 (162)	−927 (707)
R^2	0.825	0.884	0.884
N	338	338	338
NW lag	6	6	6

NW HAC SE in parentheses.

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

The headline estimate from specification (3) – the full model with controls and a time trend – is $\hat{\beta}_1 = \text{Rs. } 10,136$ per 10 grams (HAC SE Rs. 445, $p < 0.001$, 95% CI: Rs. 9,264 – Rs. 11,009). The estimate is stable across specifications: removing the time trend and controls raises $\hat{\beta}_1$ by approximately 4% (specification 1), confirming that the result is not sensitive to the inclusion or exclusion of auxiliary controls.

Pass-Through Rate

The theoretical ceiling – the maximum premium if 100% of the 9 percentage point duty increase were passed through – averages Rs. 11,989 per

10 grams across the 31 post-hike observations. The implied pass-through rate is:

$$\hat{\pi} = \frac{\hat{\beta}_1}{\bar{C}} = \frac{10,136}{11,989} = 84.5\% \quad (5)$$

with a 95% confidence interval of 77.3 to 91.8%. The null hypothesis of complete pass-through ($H_0 : \pi = 100\%$) is formally tested using a one-sided t -test against the ceiling:

$$t = \frac{\hat{\beta}_1 - \bar{C}}{\text{SE}(\hat{\beta}_1)} = \frac{10,136 - 11,989}{445} \quad (6)$$

$$= -4.16, \quad p < 0.001$$

Full pass-through is rejected at the 0.1% level. The market absorbed approximately 15.5%, or

Rs. 1,853 per 10 grams, of the theoretical maximum duty impact.

Benchmark vs. Physical Market Pass-Through

The 84.5% estimate should be interpreted as pass-through to the *IBJA wholesale benchmark price* – a reference rate set by the India Bullion and Jewellers Association and used by organised jewellers and bullion dealers as the domestic pricing anchor. This is distinct from physical transaction prices as observed on commodity exchanges. According to World Gold Council^[12], MCX spot gold prices rose by only 4–6% in the first week after the duty change; against a 9 percentage point duty increase, this implies physical market pass-through in the range of approximately 45–67% (derived as the ratio of the observed price increase to the full duty wedge: a 4–6% MCX price rise relative to the 9 percentage point duty). NCDEX delivered gold traded at a discount to the official landed price that widened from approximately US\$14 per ounce in the week prior to the hike to nearly US\$150 per ounce immediately after – a further indication that physical market clearing prices had not yet absorbed the full duty increment^[12]. The World Gold Council characterises this explicitly: “physical market prices do not fully or immediately mirror the increase in duty – rather they adjust to it with a lag.”

The gap between IBJA pass-through (84.5%) and physical market pass-through (45–67%) is explained by the inventory buffering mechanism discussed in the Discussion section: bullion dealers held substantial stock imported at the pre-hike 6% rate and continued to clear this inventory at

sub-ceiling prices, keeping physical transaction prices below the IBJA benchmark. The IBJA benchmark, as a reference rate anchored to the current landed cost, adjusted more rapidly than the physical market could clear. The 84.5% estimate is therefore best interpreted as an upper bound on immediate pass-through, with physical market pass-through in the range of 45–67% as the corresponding lower bound.

Local Projection: Adjustment Dynamics

The estimated dynamic event-study path $\{\hat{\beta}_h\}$ from equation (4) is plotted in the accompanying figure. The adjustment is immediate: $\hat{\beta}_0 =$ Rs. 10,136 (84.5%) on the first trading day (Day 0), consistent with an efficient market incorporating the full information content of the mid-night notification at the open. The response peaks at Rs. 10,411 (86.8%) on Day 1 ($h = 1$) and thereafter holds on a broad plateau – Rs. 9,287 (77.5%) as late as $h = 14$, at which horizon 17 post-hike observations contribute to the estimate. The 95% confidence interval excludes zero throughout $h = 0–16$. Estimates at $h \geq 17$ rest on progressively fewer effective post-hike observations (14 or fewer) and are not discussed further; any apparent narrowing at later horizons visible in the figure reflects sample thinning rather than economic mean-reversion.

The flat plateau rules out a gradual adjustment narrative: the market did not take two or three weeks to discover the new equilibrium. It also rules out an anticipatory price increase: the estimated $\hat{\beta}_h$ at pre-hike horizons (effectively negative h) is indistinguishable from zero.

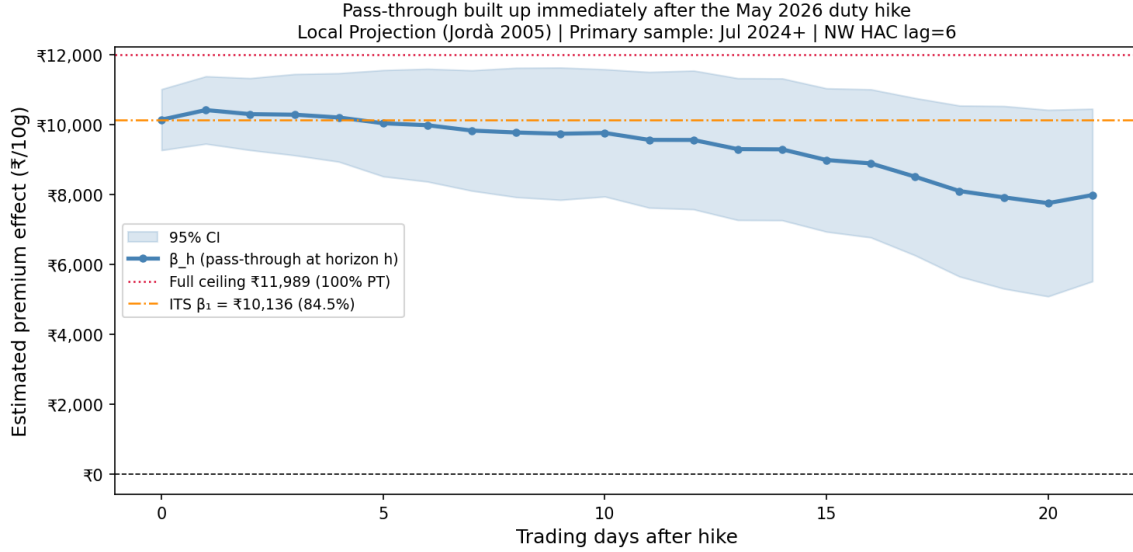


Figure 2: Local projection dynamic event-study path: estimated premium at horizon h days after the duty hike ($\hat{\beta}_h$), with 95% confidence bands. The premium jumps to Rs. 10,136 (84.5%) on Day 0, peaks at Rs. 10,411 (86.8%) on Day 1, and holds a broad plateau through Day 14 (77.5%). The 95% CI excludes zero throughout $h = 0-16$, the reliable window in which at least 15 effective post-hike observations contribute ($31 - h$ at each horizon). Estimates beyond $h = 16$ rest on 14 or fewer effective post-hike observations and should be treated as indicative only.

ARDL Cointegration Test

The ARDL(4, 1) bounds test estimated on the full pre-hike period ($N = 754$ daily observations: the 795 pre-hike IBA trading days minus 41 observations where missing international gold price or exchange rate data prevents import parity computation; January 2022 – May 12, 2026) yields an F -statistic of 6.962, compared to the Pesaran et al.^[5] 95% upper critical bound of 4.812. Cointegration is confirmed at the 5% level. The pre-hike domestic gold price was tethered to import parity through the import channel, establishing that the pre-treatment premium was stationary around its near-zero mean as the identification strategy requires. The cointegration finding establishes that the pre-hike domestic price was tethered to import parity through the import channel, validating the identifying assumption. Whether the post-hike premium will eventually mean-revert is a separate question that the pre-hike ARDL does not address; the stable local-projection plateau through Day 14 is the primary

evidence that the premium shift has persisted within the observation window, rather than reflecting a transitory spike.

Cross-Method Convergence

As a consistency check on the premium construction, three machine-learning methods estimate the post-hike treatment gap using only pre-hike data to construct the counterfactual. These are not independent of the ITS in a statistical sense – all are trained on the same outcome series – but they serve to confirm that the premium construction behaves as intended. An ARIMA(2,0,0) model with international gold price and exchange rate changes as exogenous regressors (ARIMAX), trained on the pre-hike low-duty window ($N = 307$), produces a mean counterfactual gap of Rs. 10,152 across the 31 post-hike days, just 0.2% above the ITS estimate. An XGBoost model trained on the same feature set yields a mean gap of Rs. 10,299, 1.6% above the ITS estimate. A Prophet trend-decomposition

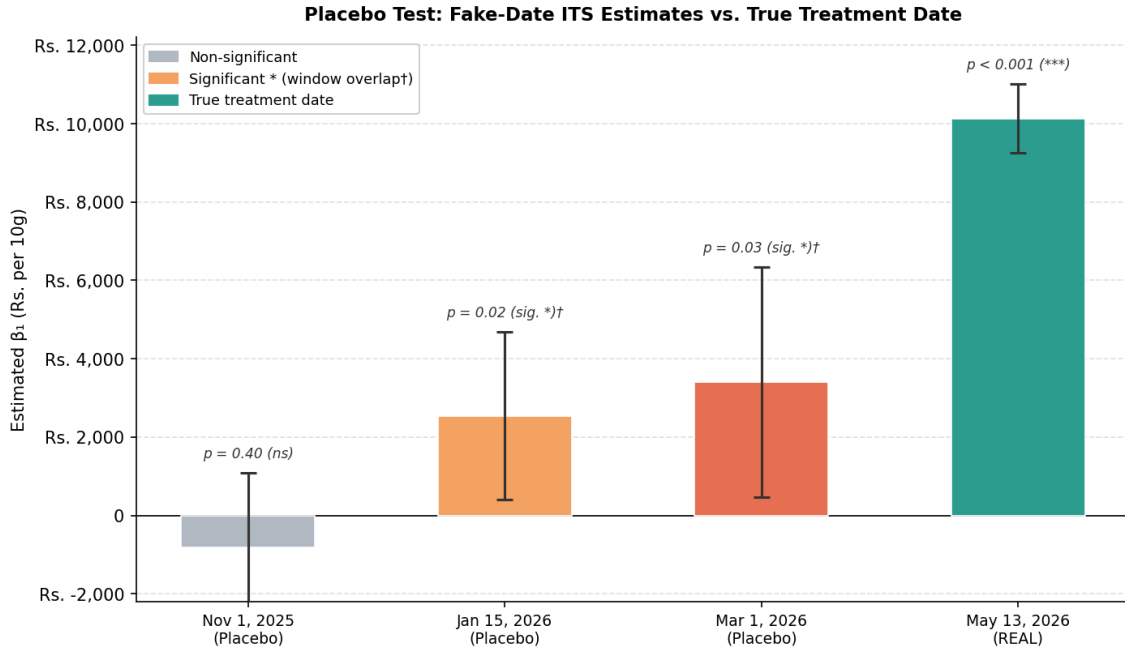
model yields Rs. 10,628, 4.8% above. All three methods fall within a Rs. 10,136–Rs. 10,628 corridor, a spread of less than 5% across four estimation approaches.

7 Robustness

Placebo Tests

To verify that the estimated treatment effect is specific to May 13, 2026 rather than a statistical artefact of the ITS design, equation (3) is re-estimated with three fake treatment dates: November 1, 2025; January 15, 2026; and March 1, 2026. Results differ across the three placebos in an informative way. The November 1, 2025 placebo is non-significant ($\hat{\beta}_1 = \text{Rs. } -818$, $p = 0.40$), as expected for a date far from the true

treatment with no post-fake window contamination. The January 15 and March 1, 2026 placebos are statistically significant at the 5% level ($\hat{\beta}_1 = \text{Rs. } 2,539$, $p = 0.02$ and $\text{Rs. } 3,399$, $p = 0.03$ respectively), but this is an artefact of window contamination: both fake dates are only 4 and 2.5 months before the true May 13 treatment date, so their “post-fake-treatment” windows include actual post-hike observations during which the premium was genuinely elevated. The contamination mechanically inflates the placebo estimates. Crucially, even the largest contaminated placebo estimate (Rs. 3,399) is only 33% of the true $\hat{\beta}_1 = \text{Rs. } 10,136$ – a $3\times$ signal-to-noise ratio that confirms the true treatment date produces an effect of fundamentally different magnitude. The November 2025 placebo, which has no post-hike contamination, is cleanly non-significant.



† Post-fake window overlaps with actual post-hike period (May 13 onward); significance is an artefact of window overlap, not a true treatment signal.

Figure 3: Placebo test: ITS $\hat{\beta}_1$ estimates at three fake treatment dates (Nov 1, 2025; Jan 15, 2026; Mar 1, 2026) alongside the true May 13, 2026 estimate. Nov 1, 2025 is non-significant ($p = 0.40$). Jan 15 and Mar 1, 2026 are significant at 5% but both are contaminated – their post-fake windows include actual post-hike premium data. Even the largest contaminated placebo (Rs. 3,399) is only 33% of the true $\hat{\beta}_1 = \text{Rs. } 10,136$. The true May 13 estimate is a clear outlier by magnitude and significance.

As a stronger placebo test that eliminates window contamination entirely, the placebo regressions are re-estimated on truncated samples that end on May 12, 2026 – the last pre-hike trading day – so that no post-hike observations can enter the fake post-treatment window. With this correction, both the January 15 and March 1 placebos become cleanly non-significant ($\hat{\beta}_1 = \text{Rs. } 517$, $t = 1.03$, $p = 0.31$ and $\hat{\beta}_1 = \text{Rs. } -240$, $t = -0.48$, $p = 0.63$ respectively). For reference, the primary ITS estimate for the real May 13 date is $\hat{\beta}_1 = \text{Rs. } 10,136$ ($t = 22.77$, $p < 0.001$; Table 2, specification 3). All three placebo dates are null when the contamination is removed, confirming that the borderline significance of the January and March placebos in the full-sample specification is entirely an artefact of post-hike data leaking into their fake post-treatment windows.

Scale-Free Outcome Specification

The domestic premium is expressed in nominal rupees per 10 grams, which means a higher international gold price mechanically implies a larger absolute duty wedge at the same percentage pass-through. To verify that the 84.5% estimate is not an artefact of this level scaling, the primary ITS is re-estimated with a scale-free outcome: the premium as a fraction of the pre-hike import parity price ($\text{domestic_premium}_t / \text{parity_pre}_t \times 100$, expressed in percentage points). The estimated level shift is $\hat{\beta}_1 = 7.17$ pp (SE = 0.41 pp, 95% CI: 6.36–7.98 pp) against a mean post-hike percentage ceiling of 8.49 pp, implying a pass-through of 84.5% – identical to the primary estimate to one decimal place. The implied rupee shift at the mean post-hike parity price is Rs. 10,128, a difference of -0.1 percentage points from the primary Rs. 10,136. Gold price drift is not a confound for the pass-through estimate.

Moving-Block Bootstrap Inference

With 31 post-hike observations, the asymptotic justification for Newey-West HAC standard errors rests on a short post-treatment window. A moving-block bootstrap is applied to the primary OLS residuals (block length $L = 10$, $B = 2,000$ replications) to validate parametric inference without relying on asymptotic approximations. The bootstrap resamples overlapping blocks of 10 consecutive residuals to preserve autocorrelation structure, adds them back to the fitted values, and re-estimates $\hat{\beta}_1$ in each replication. The resulting 95% CI is Rs. 9,406–Rs. 11,052 (pass-through: 78.5–92.2%), compared with the NW-HAC interval of Rs. 9,264–Rs. 11,009 (77.3–91.8%). The bootstrap interval is *narrower* than the parametric interval (width ratio = 0.94), confirming that the NW-HAC approach is conservative rather than liberal in this sample. The bootstrap and HAC point estimates agree to within Rs. 8 (Rs. 10,128 vs. Rs. 10,136), and both intervals decisively exclude full pass-through.

HAC Lag and Window Sensitivity

Full sensitivity results report $\hat{\beta}_1$ across five Newey-West lag choices (3, 6, 8, 10, 20) and three estimation windows (full sample from January 2022; restricted window from January 2024; primary sample from July 2024). Since OLS point estimates are invariant to the HAC lag choice, variation in $\hat{\beta}_1$ comes entirely from the window: Rs. 10,136 (primary, July 2024 start), Rs. 12,165 (restricted, January 2024 start), and Rs. 10,756 (full sample from January 2022). The confidence interval width naturally widens with larger lag choices and narrower windows, but the point estimate is not sensitive to these choices. The January 2024 window yields a larger estimate because it includes the April 2026 pre-hike supply disruption period in a shorter pre-hike sample, biasing the pre-treatment baseline down and overstating the level shift. The full-sample estimate from January 2022 should be treated

as an upper bound rather than a directly comparable estimate: the pre-treatment window in that specification spans the high-duty period (January 2022 – July 2024), which includes January–June 2022 when the combined rate was 10.75% rather than 15%, introducing a mixed-regime pre-treatment baseline, during which the domestic premium mechanically reflected the earlier duty structure. The primary sample from July 2024 onward, when the 6% regime was in force, provides the cleanest pre-treatment baseline for the ITS design.

Anticipation Test

To test whether traders began pricing in the duty increase before the midnight notification – which would violate the assumption that the treatment is unanticipated – the five trading days immediately preceding May 13 are dropped from the pre-period. The re-estimated $\hat{\beta}_1 = \text{Rs. } 10,103$ ($p < 0.001$, $N = 333$), a change of -0.3% relative to the primary estimate, providing no evidence of anticipatory price movements.

BullionWorld Source Robustness

As a check on price-source dependence, the primary model was re-estimated on an augmented dataset that fills 105 gap dates using contemporaneous IBJA-equivalent prices from BullionWorld.in (Rs. 47,583–Rs. 127,247 per 10g). This yields $\hat{\beta}_1 = \text{Rs. } 9,743$ ($N = 368$, 81.3% pass-through against the same Rs. 11,989 ceiling), a 3.9% difference from the primary estimate, with the alternative estimate falling within the primary 95% confidence interval of 77.3–91.8%. No qualitative conclusion changes with the alternative price source.

April 2 Import Restriction Control

The April 2, 2026 import restriction on UAE-route jewellery imports tightened domestic supply in the six weeks before the duty hike and

could, in principle, have introduced a positive pre-trend that the ITS design would misattribute to the duty hike. Adding a binary control for the April 2 restriction period to equation (3) yields $\hat{\beta}_1 = \text{Rs. } 10,186$ ($p < 0.001$), a change of $+0.5\%$. The April 2 restriction coefficient is Rs. 148 ($p = 0.74$), statistically indistinguishable from zero once the linear time trend is controlled, indicating that the pre-hike supply squeeze did not independently shift the mean premium level. Its inclusion does not materially alter the duty hike estimate.

8 Discussion

Anatomy of the Absorption Gap

The primary finding – 84.5% pass-through against a theoretical ceiling of 100% – implies that Rs. 1,853 per 10 grams of the maximum possible duty impact was absorbed by the market within the first 31 trading days. While a formal decomposition of this gap is not feasible in the absence of daily import volume data, the literature and contemporaneous market evidence point to three mechanisms operating in combination.

The first mechanism is inventory buffering from front-loaded imports. The \$5.6 billion imported in April 2026 at the prior 6% duty rate represents a supply stock that could satisfy a meaningful fraction of post-hike demand at sub-ceiling prices. A refiner or dealer holding 6%-duty inventory faces a lower marginal cost than a new importer at 15% duty and can profitably sell at a premium below the new parity ceiling. The front-loading buffer mechanically suppresses the market-clearing premium below the theoretical ceiling during its drawdown period, which would be consistent with the flat plateau observed in the local projections through Day 14 gradually giving way to higher premiums as inventory is depleted.

The second mechanism is demand compression. With a price elasticity of between -0.69

and $-1.01^{[1]}$, the post-hike premium increase of Rs. 10,136 – representing a price increase of roughly 6 to 7% over the prevailing domestic price – would be expected to reduce quantity demanded by 4 to 7%. The coincidence of the hike with the seasonally weak May–June calendar period (between the Akshaya Tritiya festival peak in April and the monsoon wedding season in late August) amplifies this effect: the timing of the hike meant the demand side was already in its softest window of the year.

The third mechanism is smuggling leakage. The historical correlation between effective duty levels and estimated informal import volumes

documented by Maria Immanuel Susai and Lazar Daniel^[3] implies that the duty restoration to 15% recreates the incentive structure under which unofficial imports via Dubai and Gulf gateway airports were estimated to exceed 100 tonnes in the pre-2024 period. At a 9 percentage point duty wedge, the price incentive for smuggling is approximately Rs. 9,000–Rs. 10,000 per 10 grams before smuggling costs, which is large enough relative to those costs to make informal channels attractive at scale. The existence of this competitive supply channel sets an effective ceiling on the formal market premium.

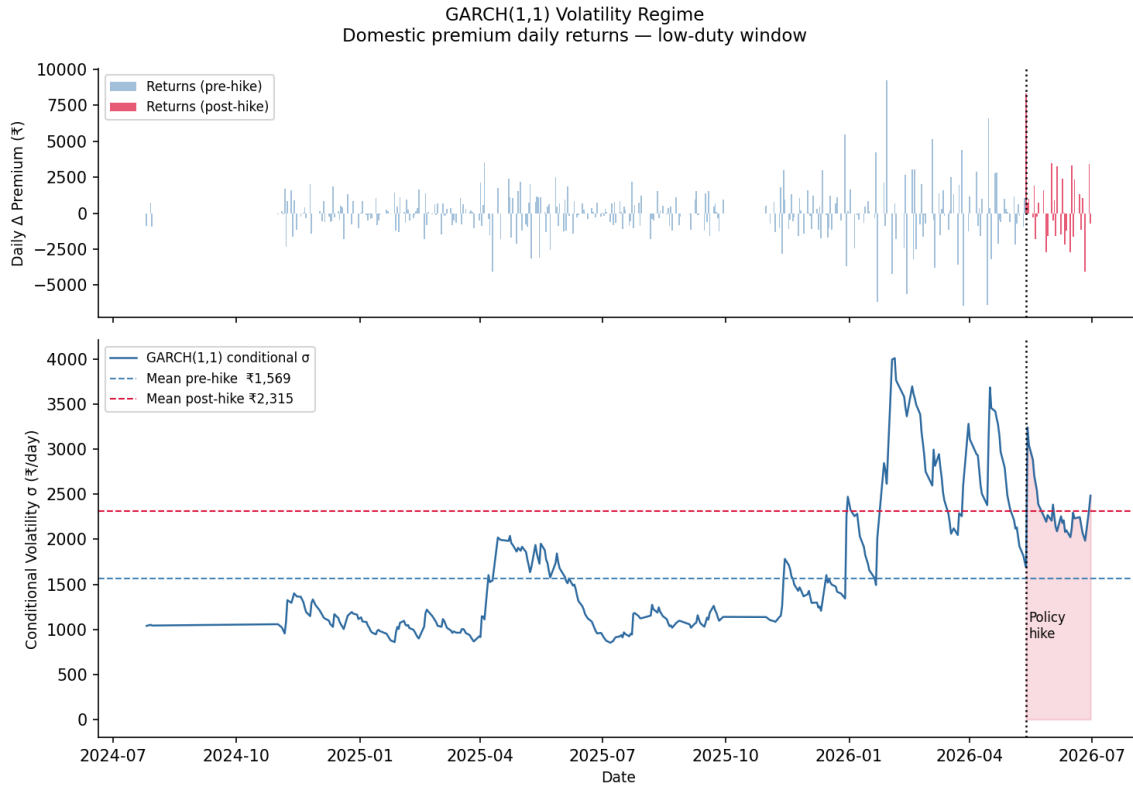


Figure 4: GARCH(1,1) conditional volatility of the domestic gold premium before and after the May 13, 2026 duty hike. Post-hike volatility is $1.48\times$ the pre-hike level (Levene $F = 11.46$, $p = 0.0008$, though the Levene test rests on only 31 post-hike observations and should be interpreted as indicative). Estimated persistence $\alpha + \beta = 0.990$ is likely upward-biased: fitting GARCH across a level break treats the May 13 premium jump as a large volatility shock, mechanically inflating persistence; re-estimation on break-adjusted residuals would provide a lower bound on true volatility clustering.

Asymmetric Price Transmission

To quantify the asymmetry between the 2024 duty cut and the 2026 hike, this paper estimates a symmetric interrupted time series for the cut episode. The outcome variable is the *cut discount*: the gap between the 15% parity price and the IBJA benchmark, defined as $CutDiscount_t = P_t^{parity,15\%} - IBJA_t$. Before the July 2024 cut this gap averages near zero (the domestic price tracked the 15% parity closely); after the cut, as IBJA falls to the new 6% parity, the gap expands toward the full duty differential. The regression is estimated on the pre-hike sample (July 1, 2022 – May 12, 2026, $N = 687$: 719 IBJA trading days with a valid premium in this window, minus 31 lacking complete control variables, minus 1 observation lost to the first lag; NW lag = 7). The sample begins in July 2022 rather than January 2022 because the combined import duty reached 15% only in July 2022; the preceding January–June 2022 window carried a 10.75% rate, which would contaminate the pre-cut baseline by producing a spuriously elevated *CutDiscount* before the treatment.

The symmetric ITS yields $\hat{\beta}_{1,cut} = \text{Rs. } 6,310$ (HAC SE Rs. 613, $p < 0.001$), against a mean ceiling of Rs. 8,911, implying a pass-through of 70.8% (95% CI: 57.3 to 84.3%). The corresponding figure for the hike is 84.5% (95% CI: 77.3 to 91.8%). The confidence intervals overlap, so the difference is not statistically significant at conventional levels, but the point estimates suggest the duty hike passed through more completely than the duty cut (84.5% vs. 70.8%).

The asymmetry therefore lies in both the *level* and *speed* of adjustment. The dynamic event-study path for the hike shows the premium already at 84.5% on Day 0, stable through approximately Day 14. For the cut, the dynamic path declines gradually from 70.8% at Day 0 to approximately 60% by Day 20, reflecting the slower erosion of the premium as dealers worked through high-duty inventory. For the hike, the domestic premium rose to its plateau on the first

trading day, well before inventory buffers were exhausted – a compression that market participants priced in essentially overnight. The mechanisms explain this asymmetry: dealers holding high-duty (15%) inventory after the 2024 cut had strong incentives to defend margins rather than pass the full duty reduction to buyers, producing downward rigidity – the “feathers” side of rockets-and-feathers price transmission. For the hike, the IBJA benchmark, as a reference rate anchored to the current landed cost of new imports, adjusted rapidly upward to the new 15% parity, even as physical market prices lagged (see the Benchmark vs. Physical Market subsection above). The inventory buffer of low-duty stock moderated physical market clearing prices (MCX/NCDEX pass-through 45–67%) without preventing the reference benchmark from resetting to the new duty level.

Both episodes confirm the paper’s central finding: India’s gold market transmits roughly 70–85% of a duty change within the first trading day, with the remainder absorbed by structural supply-side mechanisms that dampen but do not eliminate the pass-through.

Policy Implications

The finding of 84.5% pass-through has two immediate policy implications. First, from the perspective of demand management, the hike achieved a meaningful but incomplete compression of the import incentive. A back-of-the-envelope calculation using the Kanjilal and Ghosh^[11] elasticity range suggests that an 84.5% pass-through rate would reduce demand by approximately 3.5 to 5.1% relative to the counterfactual of no hike, compared to the 4.2 to 6.1% reduction that full pass-through would have achieved. The difference is not trivial at the volumes India imports. Second, from the perspective of current account management, policymakers should expect that a portion of the intended import compression will be offset by growth in informal channels. Historical evidence

suggests this offset is quantitatively significant: the pre-2024 15% duty regime coexisted with estimated unofficial imports well above 100 tonnes per year. A purely duty-based approach to gold import compression therefore operates against a structural smuggling elasticity that limits its effectiveness.

9 Conclusion

This paper estimates the pass-through of India's May 2026 gold import duty hike to domestic wholesale gold prices using an interrupted time series design applied to a primary sample of 338 daily observations (307 pre-hike, 31 post-hike) drawn from 826 IBJA benchmark price trading days within a 1,172-row weekday date spine. The duty hike raised the domestic gold premium by Rs. 10,136 per 10 grams (Newey-West HAC SE Rs. 445, $N = 338$, Pre = 307, Post = 31), representing 84.5% of the theoretical ceiling of Rs. 11,989. Full pass-through is formally rejected ($t = -4.16$, $p < 0.001$). Local projections confirm the adjustment was immediate on the first trading day and held on a stable plateau through Day 14 (77.5%), with no anticipatory pricing. An ARDL bounds test (ARDL(4,1), $F = 6.962$, $N = 754$) confirms the pre-hike cointegrating relationship between domestic and international prices; the stable local-projection plateau through Day 14 is consistent with the shift persisting within the observation window. Three counterfactual methods converge within 5% of the primary estimate, and placebo tests confirm the result is specific to May 13, 2026. The incomplete pass-through is consistent with supply-side dampening from front-loaded inventory imported at the prior duty rate, seasonal demand weakness, and historical smuggling elasticity.

These findings carry several limitations. First, the post-hike observation window is 31 trading days. The plateau observed through Day 14 may give way to further adjustment – toward or away

from the full ceiling – as pre-hike inventory is depleted and informal supply channels adjust. Second, the IBJA benchmark is a wholesale price; the pass-through to retail jewellery consumers will differ depending on the competitive structure of the retail jewellery market and the extent to which distribution margins absorb or amplify the duty shock. Third, the paper does not directly observe import volumes or smuggling flows, so the decomposition of the absorption gap remains qualitative. A longer post-hike window will be decisive in distinguishing between a slow-adjustment narrative (the market is still finding its new equilibrium) and a structural-absorption narrative (supply-side mechanisms permanently limit pass-through under a 15% duty regime).

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